ALUMINATION: An app to connect with the Alumni's of the institute

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***Abstract*—This work aims to create an alumni portal named Alumination, with distinct student and alumni interfaces, employing full-stack web development concepts to facilitate interactive discovery of alumni. The system has a Tinder-like swipe interface with ease-of-use navigation buttons, enabling students to browse alumni profiles according to hobbies, profession, and academic background. Through the use of scalable architecture with React.js, Node.js, and minimal usage of PostgreSQL, the portal is made modular, high-performance, and clean data-handling. Authenticated access provides secure interactions, and features such as search, messaging, inboxes, and announcements provide two-way communication and active alumni participation.**

***Index Terms*—Alumni dashboard, Alumni-Student Networking, Mentorship system, Event Announcements, Career Mentorship, Swipe Interface, Institutional Development.**

I. Introduction

Academic campuses usually do not have efficient platforms for impactful student-alumni interaction, particularly in areas such as mentorship, career research, and networking. Conventional systems are generally old, inflexible, and have minimal interactivity, limiting students' exposure to valuable alumni advice. This emphasizes the necessity for an up-to-date, easy-to-use, and feature-packed networking system.

To bridge this gap, Alumination is proposed—a full-stack web application engineered for personalized, interactive alumni-student interaction. Students can browse alumni profiles through a Tinder-style swipe interface, look up important information such as names, registration numbers, and employment status, and send direct messages. Alumni, through a separate portal, can search students by name, look up academic profiles, and administer assignments via an inbox-style user interface. Students can also send broadcasts, which are received by alumni through a formalised feed to foster community participation.

Technically, the application employs a React.js frontend (using Vite) and a common Express.js backend, taking advantage of in-memory JavaScript objects to facilitate quick development. Its unique feature is an Instagram-like messaging UI with threaded chat views, animated modals, and toast notifications. Profile views are simple and informative, and the swipe discovery system provides fast and smooth navigation through alumni profiles.

Alumination encourages two-way engagement that facilitates mentorship and career advice, providing both students and alumni with a responsive, accessible environment to interact. This site creates long-term academic and professional relationships across generations.

This work responds directly to these issues with the following major contributions:

1. Swipe-Based Alumni Discovery: A natural swipe interface allows students to navigate alumni profiles quickly and interactively.
2. Seperate Interfaces for Alumni and Students: Tailored frontends support the varying needs and functionalities of both user groups.
3. Strong Direct Messaging System: Two-way messaging with structured inbox views facilitates effective communication and mentorship.
4. Announcement Broadcasting Feature: Students can post event-related announcements viewable by alumni, which increases institutional connectivity.
5. Lightweight and Modular Codebase: Support for in-memory data structures facilitates agile development and provides a foundation for future integration with persistent storage systems.

By combining these aspects, this effort sets a new standard for digital alumni participation, providing an interactive, scalable, and personalized platform that fills gaps within academic communities.

The rest of this paper is organized as follows: Section II covers relevant work in alumni engagement websites and digital mentoring; Section III outlines the system architecture and design; Section IV describes implementation and user interface flows; Section V concludes with comments and sketches future development possibilities.

II. Literature Survey

Young et al. (2024) [25], underscored the necessity of tailored tracking tools in milestone-focused graduate schooling. They noted that a milestones dashboard maximizes

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transparency, improves data consistency, and aids in informed decision-making throughout decentralized programs. Wei (2024) [24], investigated the application of HOG feature extraction in the analysis of student behavior in civic education. Through the integration of a learning dashboard and implementation of K-prototype and Kruskal-Wallis algorithms, the research determined the critical factors that promote motivation, engagement, and performance in blended learning settings.

Goss and Lui (2024) [9], investigated the effects of school climate on college and career readiness in California high schools. Employing a quantitative design, the authors established that caring relationships have a positive effect on readiness, with meaningful participation having a negative correlation, with an emphasis on the importance of caring school communities. Carmichael et al. (2024) [16], probed the influence of personalized data dashboards on adherence in a 21-day EMA of drug use in college students. Though adherence was equal across groups, users finished surveys sooner and reported that messages were more compelling, implying heightened efficiency and engagement.

Bansak et al. (2024) [3], implemented student-initiated learning modules in macroeconomics to improve engagement through active research and analysis. Exercises such as policymaker profiling, monetary policy debates, and FRED data analysis enhance critical thinking, data literacy, and understanding, even though they lack quantitative assessment. Afshari et al. (2023) [2], created a computer-based student dashboard for dental patient management using electronic health records. After implementation, the dashboard resulted in decreased overdue patients and increased compliance, with increased student engagement and quality of patient care.

Ismail et al. (2022) [11], developed an exploratory dashboard to show possible and existing alumni data in Universiti Teknologi MARA through multidimensional and hierarchical approaches. The dashboard was successful in showing employment and demographic information, although privacy limitations restricted the diversity of data, with suggestions for increased data fields in future studies. Hashish et al. (2025) [1], carried out a qualitative study to explore stakeholders' views on nursing education accreditation in a Saudi college. Six main themes were identified, showing the importance of accreditation in guaranteeing quality and employability, along with uncovering issues such as administrative load and the necessity of more robust institutional backing.

Cote et al. (2025) [6], surveyed alumni attitudes towards STEM experiences, and found concerns, misconceptions, and low confidence, with 86% unsure of their future in STEM despite previous exposure. Support systems such as STEM clubs promoted belonging and resilience, with an asset-based framework informing the research on community college students. Johnsen et al. (2024) [13], assessed the effect of critical thinking practice during dental school on application by graduates. Results indicated 60% utilized these skills on a daily basis, with 53% implementing them as individual procedures—most frequently in treatment planning—confirming the worth of incorporating critical thinking within dental education.

Jewell et al. (2025) [12], investigated the incorporation of practice change learning within the PharmD curriculum at The Ohio State University. Through interviews, focus groups, and curriculum mapping, the research identified areas for enhancement—including value-based outcomes and communication simulations—with a faculty lead tasked with spearheading curricular improvements. Holden et al. (2024) [10], reviewed King's College London's rapid-track dental program for medical graduates over a span of 14 years. Of 133 graduates, 70% followed careers of dual qualification—mostly OMFS (51.6%)-with low attrition and robust preparation calling for wider UK implementation.

Forrester et al. (2025) [8], examined Dickinson College's interdisciplinary method of developing a data analytics major, focusing on faculty initiative across disciplines, institutional backing, and careful curriculum development. The program, which had to operate under resource limitations, was able to successfully combine data science, ethics, and applied learning, with future growth planned and a proposed data science center. Dominguez and Ueno (2025) [7], recognized organizational impediments to campus support services for post-foster care youth, such as shortages of resources, bureaucratic barriers, and fragmented organizational structures. These interrelated problems impede program effectiveness, underscoring the necessity for enhanced institutional coordination and focused resource allocation.

Shania et al. (2024) [21], evaluated Kanban implementation in a software firm, citing challenges including project delays, WIP management, and low understanding. The research suggests targeted training and workflow improvements to mitigate these challenges. Rajput et al. (2024) [19], created a dashboard to support pre-medical students in tracking academic and extracurricular performance. Although faculty reaction was favorable, the study identified limitations such as lacking student feedback, formal usability testing, and implementation assessment.

McCulloch et al. (2021) [17], investigated the application of visualization methods to better comprehend issues for college students with autism. The research points to how these tools can make support systems more beneficial and facilitate increased accessibility in higher education. Pereira et al. (2022) [18], examined the way in which software startup experts make use of virtual Kanban boards for workflow management, enhancing communication, and monitoring progress. Although useful, the research established that most users struggled to use the tools in a useful manner.

Sanusi et al. (2023) [20], analyzed USM graduate conversion and employability using data analysis and machine learning. The study identified low PhD and MSc conversion rates with a focus needed on policy reforms to enhance academic progression and workforce participation. Khairunizan and Feisal (2023) [15], investigated the ways in which data quality enhancement enhances alumni tracking dashboards with emphasis on cleaning issues and new methods for record management efficiency and accuracy.

Kelly et al. (2023) [14], examined the influence of social and emotional support during college on foster care alumni post-college outcomes. Positive correlations with greater happiness and lower reliance on public aid were established, but there were no significant effects on work or income. Bettinger and Fidjeland (2024) [5], investigated the correlation between college rankings, alumni satisfaction, and labor market outcomes. The results indicated that students appreciate social experiences and personal development more than employment opportunities when considering their college experience.

Wang (2024) [23], investigated the ways to raise overseas alumni contributions through a coupled model. In their research, they discovered that tax incentives, networking, and reputation drive giving, with tax benefits being the most significant driving force. Baumann and Halpern (2024) [4], examined the role of perceived value and satisfaction in influencing alumni loyalty. The research revealed that although alumni activities and institutional reputation increase perceived value, overall satisfaction is more instrumental in creating long-term loyalty.

Wang et al. (2022) [22], assessed a Learning Analytics Dashboard (LAD) embedded in the IIOE platform for online professional learning of college instructors. Participants liked and benefited from the LAD, although its effect on actual behavior change was minimal.

III. Proposed Methodology

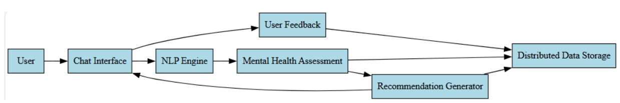

Fig. 1. System Overview of the Alumni Portal

In Fig.1., the system implemented is a full-stack web application built on modular and component-oriented architecture to enable seamless interaction between alumni and students. The system offers a swipe-based interface for finding alumni, messaging for direct communication, and separate frontend interfaces depending on user roles. Implemented in React.js, Express.js, and JavaScript-based mock data storage, the implementation aims at rapid prototyping while ensuring modularity and design simplicity.

A. System Architecture

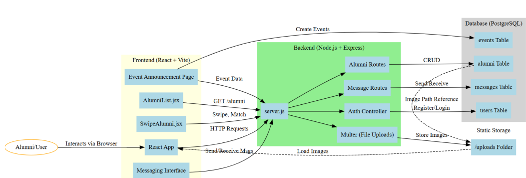

Fig. 2. Alumni Portal System Architecture

From Fig.2., the proposed architecture is organized into three primary layers that facilitate clear separation of concerns and future extensibility:

- **Frontend Layer**: The frontend is developed on top of React.js and compiled through Vite for efficient development locally. The interface allows for student interaction with alumni profiles through a Tinder-inspired swipe interface utilizing ”Ahead” and ”Change” buttons as navigators. Every profile provides the name of the individual, registration number, semester, and profile picture. The student view and alumni view are both created as separate React components to make the experience customized based on user role.
- **Backend API Layer**: Express.js is utilized to develop the backend, and it presents several RESTful endpoints for handling alumni discovery, messaging services, and handling announcements. The API layer gets and returns JSON responses between the frontend and backend, and is fueled by in-memory data structures using JavaScript objects for easy iteration and simplicity in the prototyping phase.
- **Database Layer**: At this development phase, the system does not have an integrated persistent relational database. Temporary JavaScript data objects mimic alumni and student records. This choice is done on purpose for rapid testing and future flexibility. Integration of long-term storage with PostgreSQL or equivalent technologies is scheduled for later development phases.

B. Swipe Interaction Interface

1) **Swipe Simulation**: Users browse alumni profiles with ”Ahead” to advance and ”Change” to iterate through accessible profiles. This simulates the experience of swipe-based apps with button interactions.
2) **Mock Profiles**: The entire student and alumni data are mock profiles residing in hardcoded server-side JSON objects. They consist of dummy names, images, and academic information, and represent not actual persons. Having dummy data allows for prototyping with respect to keeping the data confidential and consistent design.
3) **Modular React Components**: Every interactive element of the UI, like the swipe view, profile cards, and message forms, is wrapped in React components such as `SwipeAlumni.jsx`. These components hold state locally, take care of user events, and isolate visual logic for simplicity in maintenance.

C. Two-Interface Model

To accommodate both stakeholders of the system, two user-facing interfaces have been created:

- **User Interface (Student View)**: This view enables students to browse alumni through the swipe interaction, see in-depth profiles, start direct messaging, and post announcements to the system.
- **Alumni Interface**: The alumni interface for facing allows alumni to search student profiles by name, see profile information like registration number and semester, and enable two-way communication using an inbox interface. Alumni can also receive announcements from users.

D. Deployment and Testing

The system has been tested locally with the following configuration:

- The React frontend is served on Vite at `localhost:3000`.
- The Express.js backend server runs on `localhost:5000`.
- Mock APIs are tested in Postman for route and payload format validation.
- Profile pictures and other static files are served from a static directory mapped using Express middleware.

E. Evaluation Metrics

The system will be measured against the following: The performance of the system will be measured using the following metrics:

- **Interface Responsiveness**: Performance of swiping and messaging interface speed and smoothness.
- **API Query Performance**: Backend data endpoint response time and reliability.
- **User Satisfaction**: Informed user feedback gathered during testing interactions.
- **Maintainability**: Ease of introduction and testing of new functionality like secure login, event calendars, and data persistence.

IV. Results and Discussion

The alumni-user interaction platform is a web application that is full-stack and implemented using React.js for both User and Alumni interfaces, with a common Express.js backend. It enables profile discovery, two-way messaging, and announcement delivery through both interfaces. All data—users, alumni, messages, and announcements—are handled through in-memory arrays to enable quick prototyping and demonstration. Below is the frontend demonstration of the User and Alumni interfaces:

1) User Interface:

- **Alumni Discovery**: Students can view Alumni through a Tinder-like swipe interface or locate individuals on a search-by-name page. Navigating in the swipe interface is facilitated by the "Ahead" button, which goes to the next profile, and the "Change" button, which rotates through other profiles for wider discovery.

  Fig.3. shows how the Tinder-like swipe interface looks:

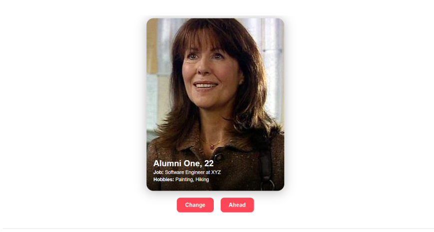

Fig. 3. Tinder-like swipe interface

Fig.4. shows how the Search bar looks:

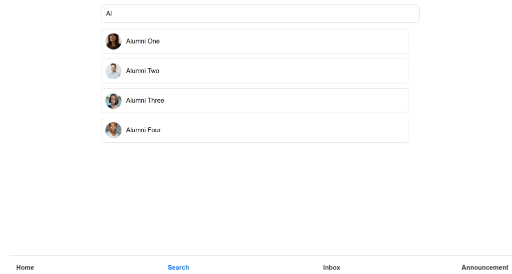

Fig. 4. Search bar interface

- **Profile Viewing**: The alumni profile shows major information such as name, city, interests, accomplishments, LinkedIn profile, and a picture. A separate profile view option enables users to look at this information prior to making contact which is shown in Fig.5.

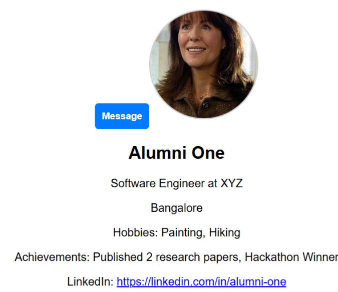

Fig. 5. Sample Alumni Profile

- **Messaging System**: A floating button on alumni profiles to send a message opens an animated modal with autofocus activated for easy input. After sending a message, a toast-style notification offers direct feedback which is shown in Fig.6.

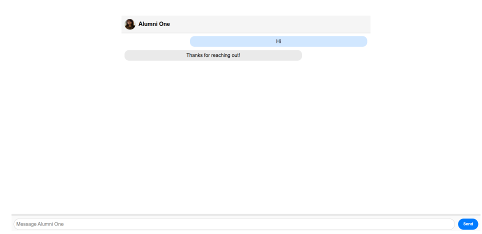

Fig. 6. Direct Message interface with Alumni

- **Inbox**: A list of everyone the user has chatted with appears, with profile pictures and names as shown in Fig.7.
- **Give Announcement Page**: Students are able to post announcements, like event invites, fest information, or club events, via a special page that are forwarded to the alumni interface and are stored on the backend for display. This interface is shown in Fig.8.

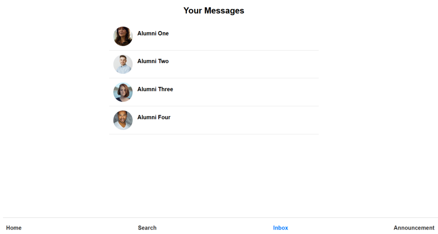

Fig. 7. User Inbox

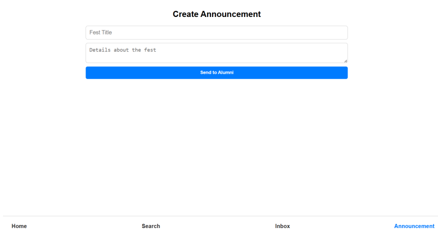

Fig. 8. Post Announcement Page

2) *Alumni Interface:*

- **Receiving Announcements**: Alumni are sent announcements from the user side, for example, events in the future or educational opportunities. The page has a scrollable list of announcements, and highlighted advertisement-style banners are located at the top of the page to highlight major notices or reminders so that alumni do not miss out on important information as shown in Fig.9.

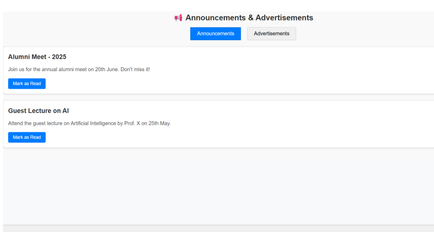

Fig. 9. Receive Announcement Page

- **User Discovery**: Students can be searched for by name with live filtering. Clicking on a result opens a full student profile as shown in Fig.10.

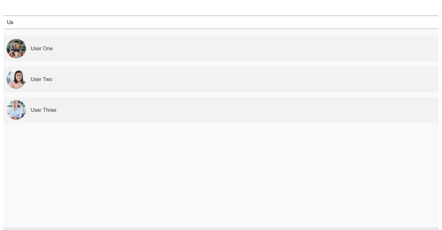

Fig. 10. Search Bar for User Discovery

- **Profile Viewing**: Full name, standardized registration number (BL.EN.U4XXX2XXX format), semester of study, and profile image comprise student profiles as shown in Fig.11.

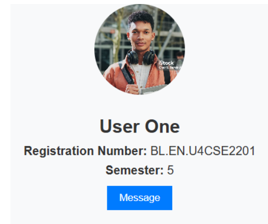

Fig. 11. Sample User Profile

- **Messaging Users**: Alumni are able to start and follow through with dialogues with students. Messages remain locally and are displayed in an alumni-specific inbox with uniform layout and styling as shown in Fig.12.

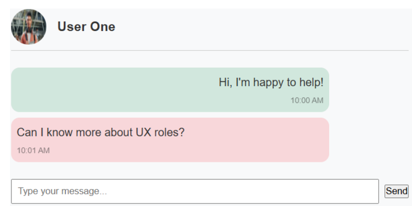

Fig. 12. Direct Message interface with User

- **Inbox**: A list of everyone the Alumni has chatted with appears, with profile pictures and names as shown in Fig.13.

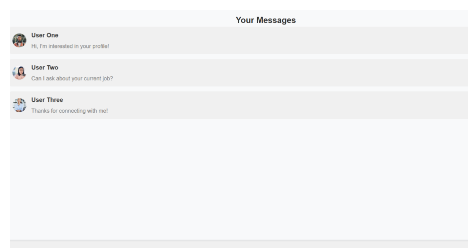

Fig. 13. Alumni Inbox

- **Help Page**: The Help page grants alumni unique access to the contact information of five major university officials. This functionality is made to make direct contact more possible for help—either in the form of regular assistance or emergencies—so that alumni are able to reach the right staff when necessary as shown in Fig.14.

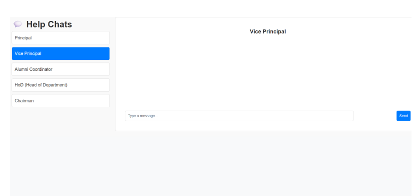

Fig. 14. Help Page

Thus the overall interface demo showcases the major features and usability of the platform. It ensures effective two-way communication and profile interaction between alumni and students. The design continues to stay modular, responsive, and user-friendly. In spite of prototype constraints such as in-memory data storage and no complete authentication, the system does an excellent job of demonstrating its idea. Overall, it provides a solid foundation for further refinement.

V. Conclusion and Future Scope

Alumination is a dynamic platform that allows students to have meaningful interactions with alumni in terms of mentorship, networking, and career advice. The student interface is React.js-powered with a PostgreSQL database as the backend, providing a structured data handling and responsiveness. Coupled with the shared Node.js backend, the platform provides a seamless, swipe-based discovery, clean profile display, and two-way messaging, which renders engagement easy and convenient. By making alumni-student relations simpler, Alumination enhances institutional ties and school spirit. Future plans will add real-time messaging, push notifications, OTP-based login, and deeper integration with institutional systems—improving security, scalability, and long-term user adoption.

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